RESEARCH PROPOSAL

Smart E-Commerce Purchase Intent & Conversion Predictor

A leakage-safe framework for prediction, evaluation, and timely intervention.

SUBMITTED BY

Sahar Osama

Sahar Reda

Samah Mohamed

02 / PROJECT SCOPE

What the project covers

The project focuses on predicting whether an e-commerce website session will result in a purchase, using session-level browsing and engagement features from a public dataset such as the UCI Online Shoppers Purchasing Intention Dataset. The final dataset, license, and feature availability will be verified before modeling.

IN SCOPE

Data understanding, cleaning, feature-timing audit, and exploratory analysis.

MODEL BENCHMARK

Logistic Regression, KNN, SVM, Decision Tree, and Random Forest.

DELIVERABLE

A fair model comparison and a practical score-to-action demonstration.

03 / PROBLEM & SOLUTION

From uncertain browsing to timely action

THE PROBLEM

Most sessions do not end in a purchase. Marketing teams may struggle to distinguish visitors with strong purchase intent from casual browsers. Class imbalance, data leakage, and features available only after a session can also produce misleading results.

THE SOLUTION

Build and compare five standard classifiers using a stratified split-first workflow. Fit preprocessing and any balancing on training folds only, evaluate on an untouched test set, and use F1-Score plus ROC-AUC to guide a proportionate, targeted intervention.

04 / PROJECT OBJECTIVES

Measurable objectives

01 **Establish a leakage-safe pipeline**
Split data before balancing; fit preprocessing only within training folds.

02 **Train five comparable models**
Use consistent splits, cross-validation, and reproducible settings.

03 **Evaluate performance fairly**
Compare F1-Score and ROC-AUC, supported by precision, recall, and confusion matrices.

04 **Translate scores into action**
Define a clear intervention rule while communicating uncertainty and limitations.

05 / COMPETITORS & ALTERNATIVES

Where this project fits

ALTERNATIVE	MAIN CAPABILITY	PROJECT DIFFERENTIATION
Google Analytics 4	Tracks user journeys and conversion events.	Controlled, leakage-safe comparison of predictive classifiers.
Shopify Analytics	Store-level sales and customer reports for Shopify merchants.	Platform-independent predictive study, not a store reporting suite.
Adobe Analytics	Advanced digital analytics and customer journey insights.	Transparent model evaluation and reproducible student project.
Dynamic Yield	Personalization and targeted customer experiences.	Purchase-intent scoring prototype, not production-grade personalization.

01 / OVERVIEW

Executive summary

Most e-commerce sessions end without a purchase, even when the visitor's browsing suggests real interest. Marketing teams usually cannot tell which of these sessions are worth acting on, so interventions are either applied to everyone (costly and intrusive) or applied too late (after the visitor has left).

We propose to build and rigorously evaluate a classifier that estimates, from in-session behavior, whether a visit will end in a purchase, and to turn its output into a practical signal for proactive, targeted marketing. Five standard models (Logistic Regression, KNN, SVM, Decision Tree and Random Forest) will be compared under one identical, **leakage-safe** protocol in which the data is split *before* any balancing is applied. Performance will be reported with both **F1-Score** and **ROC-AUC**, so no model is judged on a single number.

PROBLEM	TARGET USERS	APPROACH	OUTCOME
Low conversion (around 15%) leaves most sessions unresolved and high-intent visitors unseen.	E-commerce marketing, growth and product teams.	Capture, estimate, intervene: five models, one leakage-safe protocol.	A credible, comparable model ranking and a score-to-action signal.

02 / THE PROBLEM

What is going wrong today

The business problem. At a typical conversion rate of about 15%, roughly 85 of every 100 sessions end with no purchase. Among those are visitors who browsed seriously, compared products or added items to a cart. Today that intent is effectively invisible: there is no reliable way to separate "about to buy" from "just looking" while the session is still live.

The technical problem. Predicting purchase intent sounds simple, but three pitfalls routinely make published results look better than they will perform in production:

- **Class imbalance.** Buyers are the minority, so accuracy is misleading and a model can look strong while missing most purchasers.
- **Data leakage.** Balancing the data (for example oversampling) before splitting lets duplicated or synthetic rows reach the test set and inflates every metric.
- **Timing leakage.** Features that are only known at the end of a session cannot support a real-time decision, however predictive they look offline.

Problem statement

How reliably can in-session browsing behavior predict purchase, which of five standard classifiers does it best under a leakage-safe protocol, and can the result support timely, targeted intervention without overstating certainty?

03 / TARGET USERS

Who this is for

The project is designed around the people who must decide, in the moment, whether a visitor deserves an intervention, and the people who must trust the model enough to deploy it.

User	What they need	How the project helps
Marketing and CRM teams (primary)	To know which live visitors are likely buyers, so offers go to the right people at the right time.	Provides a purchase-intent score and a documented rule for turning it into an intervention.
Growth and conversion analysts	A fair way to choose between models and to judge how much to trust a score.	Delivers a like-for-like comparison with F1-Score, ROC-AUC and fold-to-fold variability.
Product and engineering teams	A model that can run on signals available during the session.	Audits every feature for availability at prediction time and keeps the pipeline reproducible.
Shoppers (indirect)	Help that is relevant and well timed, rather than blanket discounts and pop-ups.	Targeting only high-intent visitors reduces unnecessary interruption.
Students and researchers	A clear example of leakage-safe evaluation on an imbalanced problem.	Documents the split-first protocol so results can be reproduced and critiqued.

04 / WHY IT MATTERS

The value of getting this right

Consider 10,000 sessions at a 15% conversion rate: about 1,500 end in a purchase and 8,500 do not. Even a small share of those 8,500 that can be recognised as high-intent and nudged in time represents revenue that is currently lost without anyone noticing.

Benefit	Why it matters
Recovered revenue	Acting while the visitor is still on the site, rather than after they leave, targets the narrow window where intervention can still change the outcome.
Efficient marketing spend	Discounts, reminders and chat prompts go to visitors who are likely to respond instead of to everyone, protecting margin.
Better customer experience	Fewer irrelevant interruptions for casual browsers and more useful help for serious ones.
Credible evidence	Splitting before balancing and reporting two complementary metrics means the results can be trusted outside the notebook.
Honest uncertainty	Predictions are treated as probabilities that guide action, not as certainties, which keeps the intervention policy proportionate.

05 / OBJECTIVES

What we will deliver and ask

The project has four objectives, each tied to a verifiable deliverable:

#	Objective	Evidence of completion
1	Build a leakage-safe experimental pipeline in which the train/test split precedes all balancing and preprocessing.	Written leakage checklist, passed by an independent reviewer.
2	Train and tune five models (Logistic Regression, KNN, SVM, Decision Tree, Random Forest) under identical conditions.	Reproducible code with fixed seeds and shared splits.
3	Identify the strongest model using F1-Score and ROC-AUC together.	Comparison table with mean and spread across folds, plus a final held-out test result.
4	Translate the best model's output into a practical intervention signal.	Documented score-to-action policy and a working demonstration.

Research questions

Theme	Question
Reliability	Which model identifies likely purchasers most reliably?
Comparison	How does performance change across competing algorithms?
Usefulness	Can predictions support timely, targeted intervention without overstating certainty?

06 / METHODOLOGY

How we will do it

The system follows three stages: **capture** behavioral signals from the session, **estimate** purchase intent with a trained classifier, and **intervene** for high-value visitors. The experimental work focuses on the middle stage and on proving it is evaluated honestly.

Capture session signals	Split first (stratified)	Preprocess and balance (train only)	Train and tune 5 models	Evaluate on held-out test	Intervene on high intent

Leakage-safe protocol

- **Define the target and audit features.** Keep only signals that would be available while the session is live; document any feature excluded for timing reasons.
- **Split first.** Create a stratified train/test split before any other step. The test set keeps the original class ratio and is not touched until the final evaluation.
- **Fit everything on training data only.** Imputation, scaling, encoding and class balancing (oversampling or class weights) are fitted inside each training fold of a cross-validation pipeline, never on the full dataset.

- **Tune with cross-validation.** Hyperparameters are chosen on the training set only, using the same stratified folds for every model.
- **Evaluate once.** Each tuned model is scored a single time on the untouched test set; fixed random seeds make every run reproducible.

Data

We will use a session-level e-commerce behavioral dataset with a binary purchase label and a conversion rate near 15%. A public dataset such as the UCI Online Shoppers Purchasing Intention data is a natural candidate; the final choice, its license and its feature timing will be confirmed in week 1 before modeling begins.

Model benchmark

Model	Role in the study	Main settings to tune
Logistic Regression	Interpretable baseline for purchase classification.	Regularization strength and type
KNN	Tests neighborhood-based decision behavior.	Number of neighbors, distance weighting
SVM	Tests an alternative separating boundary.	Kernel, C, gamma
Decision Tree	Tests a rule-based decision structure.	Maximum depth, minimum leaf size
Random Forest	Tests ensemble behavior across multiple trees.	Number of trees, depth, features per split

07 / EVALUATION

How we will judge success

Metric	What it captures	Why we use it
F1-Score	Balance between precision and recall for the positive (purchase) class.	Shows whether a model finds buyers without flooding marketing with false alarms.
ROC-AUC	Ranking and discrimination across all classification thresholds.	Shows how well scores separate buyers from non-buyers regardless of the cut-off chosen.

Both metrics are always reported together. As supporting evidence we will add precision, recall, the confusion matrix and a threshold analysis, because the right cut-off for an intervention depends on its cost.

Success criteria

- The leakage checklist is complete and independently verified.
- All five models are compared under identical splits, preprocessing and tuning budget.
- The strongest model is selected on F1-Score and ROC-AUC jointly, and fold-to-fold variation is reported so small differences are not over-interpreted.
- The selected model is shown to improve on a majority-class baseline and on the Logistic Regression baseline; numeric targets will be fixed after the first baseline run.
- A score-to-action policy is documented, including the limits of what a probability can claim.

08 / PROJECT PLAN

Eight weeks from scoping to delivery

Phase	Lead	W1	W2	W3	W4	W5	W6	W7	W8
Scoping, dataset choice, feature-timing audit	Sahar O.								
Exploratory analysis and cleaning	Sahar O.								
Leakage-safe pipeline and checklist	Sahar O.								
Model training and tuning (five models)	All								
Evaluation framework and model comparison	Sahar R.								
Intervention policy and demonstration	Samah								
Report, slides and final presentation	Samah								

Bar colors follow the lead for each phase; “All” marks shared work.

Milestones and deliverables

When	Milestone	Deliverable
End of week 2	Data understood	Dataset note, feature-timing audit, EDA summary
End of week 3	Pipeline frozen	Leakage-safe pipeline and signed-off checklist
End of week 5	Models tuned	Five tuned models with cross-validation results
End of week 6	Comparison complete	Final F1-Score and ROC-AUC comparison with test results
End of week 8	Project delivered	Intervention policy, demonstration, final report and presentation

09 / TEAM ROLES

Three leads, equal weight

Each member leads one area of the project and also owns part of the model benchmark, so the workload is balanced across data, modeling and delivery. No one is responsible for reviewing their own work: every area is cross-reviewed by a teammate.

Member	Role	Core responsibilities	Models owned	Reviewed by
Sahar Osama	Data and Pipeline Lead	Dataset selection, cleaning, feature-timing audit, stratified split, preprocessing and balancing inside the cross-validation pipeline, leakage checklist.	Logistic Regression, KNN	Sahar Reda
Sahar Reda	Modeling and Evaluation Lead	Shared tuning framework, F1-Score and ROC-AUC computation, comparison tables and curves, threshold analysis, statistical sanity checks.	SVM, Random Forest	Samah Mohamed
Samah Mohamed	Integration and Delivery Lead	Score-to-action policy, intervention demonstration, final report and presentation, schedule and meeting coordination.	Decision Tree	Sahar Osama

Shared working agreements

- **Equal contribution.** Roughly one third of total effort each; all three take part in tuning, interpretation and writing.
- **One repository, one protocol.** All code lives in a shared repository and every model runs through the same pipeline and splits.
- **Weekly 30-minute sync.** The facilitator rotates each week, and decisions and open issues are written down.
- **Report authorship.** Each member writes the sections that match their role, and one section is co-written by all three.
- **Cover for gaps.** If someone is blocked or unavailable, the reviewer for that area steps in so the schedule holds.

10 / RISKS

What could go wrong and how we respond

Risk	Impact	Mitigation
Hidden leakage (balancing or fitting before the split, or session-end features)	Inflated metrics that do not hold in practice	Split first, fit all transformers inside training folds, run the leakage checklist and independent review.
Strong class imbalance	Models ignore the buyer class	Stratified splits, class weights or train-only oversampling, and F1-Score alongside ROC-AUC.
Overfitting during tuning	Optimistic model ranking	Cross-validation on the training set only and a single final test evaluation.
Dataset does not match live behavior	Limited real-world transfer	State the limitation clearly, avoid unavailable features, and frame results as a benchmark.
Over-trusting predictions	Poorly judged interventions	Report spread and thresholds, treat outputs as probabilities, and match action intensity to confidence.
Privacy and fairness	Intrusive or biased targeting	Use only session-level behavioral signals, no personal identifiers, and review features for proxies.
Schedule slippage	Missed delivery	Weekly check-ins, early pipeline freeze in week 3, and reviewer cover.

11 / EXPECTED OUTCOMES

What success looks like

Rigor	Evidence	Action
A leakage-safe experimental pipeline and a fair comparative model evaluation.	The strongest model identified using F1-Score and ROC-AUC together, with uncertainty reported.	A practical signal that lets e-commerce marketing act proactively on high-intent visitors.

> *Success means credible prediction that can guide action.*